

Lecture 9 — Django REST Framework & Advanced ORM

Question Bank

Lecture 9 preparation document with contextual multiple choice questions, fill in the blanks, and answer key.

Part 1 — Multiple Choice Questions

1. Which field should be used for money values?

- A) FloatField
- B) IntegerField
- C) DecimalField
- D) TextField

2. Why is FloatField bad for money?

- A) Too slow
- B) Loses precision
- C) Cannot store decimals
- D) Cannot filter

3. What does blank=True affect?

- A) Database storage
- B) Validation/forms/API
- C) Routing
- D) SQL indexes

4. What does null=True affect?

- A) Database storage
- B) Templates
- C) Routing
- D) JSON serialization

5. Which relationship represents many products → one category?

- A) OneToOneField
- B) ManyToManyField
- C) ForeignKey
- D) JSONField

6. What does related_name='products' allow?

- A) Product.category_set
- B) cat.products.all()
- C) Product.related()
- D) category.products()

7. What does ManyToManyField automatically create?

- A) Serializer
- B) URL route
- C) Join/intermediate table
- D) ViewSet

8. What does auto_now_add=True do?

- A) Updates on every save
- B) Sets value once on INSERT
- C) Creates migration
- D) Makes field editable

9. What does auto_now=True do?

- A) Updates on every save
- B) Sets value only once
- C) Creates index
- D) Deletes old values

10. What does ordering = ['-created_at'] do?

- A) Ascending sort
- B) Random order
- C) Default descending order
- D) Deletes old objects

11. What does unique_together create?

- A) ForeignKey
- B) Composite uniqueness constraint
- C) Join table
- D) Serializer validation only

12. What does filter() return?

- A) Single object
- B) QuerySet
- C) Dict
- D) JSON

13. What does get() return?

- A) QuerySet
- B) One object or exception
- C) List
- D) Response

14. What does Q(a) | Q(b) mean?

- A) AND

- B) OR
- C) NOT
- D) XOR

15. What does $\sim Q(\dots)$ mean?

- A) AND
- B) OR
- C) NOT
- D) QuerySet

16. What does `aggregate()` return?

- A) Computed field per object
- B) One value for whole QuerySet
- C) One serializer
- D) Route list

17. What does `annotate()` do?

- A) Deletes rows
- B) Adds computed field per object
- C) Creates migration
- D) Generates serializer

18. What replaces manual `to_json()` in DRF?

- A) JsonResponse
- B) ModelSerializer
- C) HttpResponse
- D) Router

19. What does `ModelViewSet` automatically provide?

- A) Only GET
- B) CRUD actions
- C) Only POST
- D) Only serializers

20. Why is `many=True` required?

- A) For one object
- B) For QuerySet/list serialization
- C) For DELETE request
- D) For migrations

21. What does `DefaultRouter` do?

- A) Creates migrations
- B) Auto-generates URLs
- C) Creates serializers

D) Creates database

22. What does @action(detail=True) mean?

A) URL contains object id

B) Creates DB index

C) Makes serializer read-only

D) Enables pagination

23. What does fields='__all__' mean?

A) No fields

B) Only id

C) Include all model fields

D) Exclude all fields

24. What does readonly_fields do in ModelAdmin?

A) Deletes fields

B) Makes fields non-editable in admin

C) Creates indexes

D) Adds serializers

25. When does SQL execute for QuerySets?

A) Immediately

B) Only on evaluation

C) During migration

D) During serializer creation

Part 2 — Fill in the Blanks

1. Money field type

price = models._____(max_digits=10, decimal_places=2)

2. Validation/forms option

description = models.TextField(_____=True)

3. Database NULL option

description = models.TextField(_____=True)

4. Many-to-one relationship

category = models._____(Category, on_delete=models.CASCADE)

5. Reverse access

related_name='_____'

6. One-to-one relationship

profile = models._____(User, on_delete=models.CASCADE)

7. Many-to-many relationship

tags = models._____(Tag, related_name='products')

8. Created timestamp once

created_at = models.DateTimeField(_____=True)

9. Updated timestamp every save

updated_at = models.DateTimeField(_____=True)

10. Descending ordering

ordering = ['_____created_at']

11. ORM OR query

Q(price__lt=500) ____ Q(name__icontains='budget')

12. ORM AND query

Q(is_active=True) ____ Q(price__gt=100)

13. ORM NOT query

Product.objects.filter(____Q(category__name='Archived'))

14. One value for QuerySet

Product.objects._____(Avg('price'))

15. Computed field per object

Category.objects._____(product_count=Count('products'))

16. DRF serializer base class

class ProductSerializer(serializers._____):

17. Include all serializer fields

fields = '_____'

18. DRF CRUD ViewSet

class ProductViewSet(viewsets._____):

19. Serialize QuerySet

serializer = ProductSerializer(qs, _____=True)

20. Router class

router = _____()

Answer Key

MULTIPLE CHOICE

1. C
2. B
3. B
4. A
5. C
6. B
7. C
8. B
9. A
10. C
11. B
12. B
13. B
14. B
15. C
16. B
17. B
18. B
19. B
20. B
21. B
22. A
23. C
24. B
25. B

FILL IN THE BLANKS

1. DecimalField
2. blank
3. null
4. ForeignKey
5. products
6. OneToOneField
7. ManyToManyField
8. auto_now_add
9. auto_now
10. -
11. |
12. &
13. ~
14. aggregate
15. annotate
16. ModelSerializer
17. __all__
18. ModelViewSet
19. many
20. DefaultRouter